David W. Hogg

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Education

PhD 1998, Physics, California Institute of Technology.
SB 1992 (Physics), Massachusetts Institute of Technology.

Positions

current: Professor of Physics and Data Science, New York University, 2014–present.
Associate Professor of Physics with tenure, New York University, 2007–2014.
Assistant Professor of Physics, New York University, 2001–2007.
Long-term member, Institute for Advanced Study, 1997–2001.

Short-term and part-time positions

current: Senior Research Scientist, Center for Computational Astrophysics, Flatiron Institute, New York, 2023–present.
current: Adjunct Scientist, Max-Planck-Institut für Astronomie, Heidelberg, Germany, 2024–present.
Adjunct Senior Staff Scientist, Max-Planck-Institut für Astronomie, Heidelberg, Germany, 2012–2024
Group Leader, Astronomical Data Group, Center for Computational Astrophysics, Flatiron Institute, New York, 2017–2023.
Consultant, Flatiron Institute, New York, 2016–2017.
Consultant, Simons Center for Data Analysis, Simons Foundation, New York, 2015–2016.
Visiting Scientist, Max-Planck-Institut für Astronomie, Heidelberg, Germany, 2006–2012.
Visiting Professor, Department of Astronomy and Astrophysics, Columbia University, 2008
Consultant, Google Inc., 2008.
Scholar in Residence, *Spitzer* Science Center, California Institute of Technology, 2006
Visiting Professor, Department of Physics, Massachusetts Institute of Technology, 2005
Lecturer (part-time), Department of Physics, Princeton University, 1998–2001.

Administrative roles

Executive Director, *Moore–Sloan Data Science Environment at NYU*, 2013–2015.
Deputy Director, New York University Center for Data Science, 2014–2015.
Director of Undergraduate Studies, Department of Physics, New York University, 2008–2015.

Service

Chair, *Sloan Digital Sky Survey V* Advisory Council, 2023–present.
Terra Hunting Experiment Board, 2019–present.
Sloan Digital Sky Survey V Technical Advisory Group, 2017–present.
Sloan Digital Sky Survey IV Collaboration Council, 2013–2022.
Spitzer Science Center Oversight Committee, 2008–2019.
Regular panel reviewer for the US National Science Foundation and the US National Aeronautics and Space Administration.
Founder and co-organizer of hands-on research and education workshops, including *AAS Hack Together Day* (annually in January, 2013–2020), *AstroHackWeek* (five events, 2014–2017, 2020) *Preparing for TESS* (2018), *Telluric Line Hack Week* (2019), and the

Gaia Sprint (six week-long events, 2016–2022).

White House appointee to the US national Astronomy and Astrophysics Advisory Committee, 2014–2017.

NASA Extragalactic Database Users Committee, 2006–2013.

Sloan Digital Sky Survey III Collaboration Council, 2008–2011.

Panel Chair, *Spitzer Space Telescope* Time Allocation Committee, 2005.

Sloan Digital Sky Survey Collaboration Council, 1999–2004.

Leader, *Sloan Digital Sky Survey* Calibration Task Force, 2000–2003.

National Optical Astronomy Observatories Time Allocation Committee, 2000–2002.

Hubble Space Telescope Time Allocation Committee, 1999.

Honors

New York University "Golden Dozen" Teaching Award, 2025.

Neural Information Processing Systems Machine Learning and the Physical Sciences Workshop "Best 'AI for Physics' Paper Award" (first author Gregory, W. G.), 2024.

International Conference on Machine Learning Latinx in AI Workshop "Best Poster Award" (first author Tardugno Poleo, V.), 2024.

New York University "Golden Dozen" Teaching Award, 2004.

Princeton University Engineering Council Teaching Award, 2000.

Caltech Undergraduate Teaching Award, Associated Students of Caltech, 1996.

J. S. Stemple Memorial Prize, Caltech, for Physics PhD oral candidacy exam, 1995.

Phi Beta Kappa, 1992. Sigma Xi, 1992.

Award of merit, International Physics Olympiad, Bad Ischl, Austria, 1988.

Grants

ONR grant (Recommended for funding; Hogg, PI) *Mathematically principled machine learning approaches to ocean dynamics*, 1,077,458 USD, 2026–2029.

Simons Foundation Society of Fellows Junior Fellowship grant (Hogg, PI; for A. Beane), 300,000 USD (approx), 2025–2028.

NASA Hubble Fellowship grant (HST-HF2-51529.001-A; Hogg, PI; for T. Dharmawardena), *Towards a Comprehensive 3D View of Dust in the Galactic neighbourhood*, 329,000 USD (approx), 2023–2026.

NASA Hubble Fellowship grant (HST-HF2-51466.001-A; Hogg, PI; for S. Pearson), *Decoding Extragalactic Stellar Streams: What will they teach us about dark matter?*, 356,869 USD, 2021–2024.

NASA FINESST Graduate Fellowship grant (19-ASTRO20-0076; Hogg, PI; for K. Storey-Fisher), *Clustering Analysis Methods for the Next Generation of Large-Scale Structure Surveys* 135,000 USD, 2020–2024.

NASA *TESS* Guest Investigator grant (Angus, PI), *Measuring long rotation periods from TESS's short light curves*, 50,000 USD, 2019–2021.

NASA *TESS* Guest Investigator grant (Hogg, PI; for B. S. J. Pope), *Halo Photometry of Naked-Eye Stars with TESS*, 50,000 USD, 2019–2021.

NASA Astrophysics Data Analysis grant (80NSSC19K0533; Bean, PI), *Improving the sensitivity of radial velocity spectrographs with data-driven techniques*, 308,326 USD, 2019–2021.

NASA JPL Sagan Fellowship grant (1592259; Hogg, PI; for B. S. J. Pope), *Finding Planets around Naked-Eye Stars*, 352,064 USD, 2017–2020.

NSF Cyberinfrastructure for Emerging Science and Engineering Research grant (OAC-1841594; Hogg, PI), *Collaborative Research: Community Planning for Scalable Cyberinfrastructure to Support Multi-Messenger Astrophysics*, 36,469 USD, 2018–2019.

NASA SAO Einstein Fellowship grant (PF6-170154; Hogg, PI; for B. Leistedt), *Testing Fundamental Physics with Galaxy Surveys*, 342,129 USD, 2016–2019.

NSF Astronomy and Astrophysics Research grant (AST-1517237; Hogg, PI), *New Probabilistic Methods for Observational Cosmology*, 328,312 USD, 2015–2019.

Moore Foundation and Sloan Foundation joint grant (LeCun, PI), *The Moore-Sloan Data Science Environment at NYU*, 12,600,000 USD (approx), 2013–2018.

NASA *K2* Guest Observer grant (NNX16AC70G; Hogg, PI), *Ultra-precise photometry in*

crowded fields: A self-calibration approach, 100,000 USD, 2016–2017.

NASA Astrophysics Data Analysis grant (NNX12AI50G; Hogg, PI), *The Lives and Deaths of Planets and Stars in the Value-Added UV Photon Catalog*, 473,705 USD, 2012–2017.

Simons Foundation Society of Fellows Junior Fellowship grant (Hogg, PI; for B. Leistedt), 90,000 USD (approx), 2015–2016.

NSF Cyber-Enabled Discovery Type I Grant (IIS-1124794; Hogg, PI), *A Unified Probabilistic Model of Astronomical Imaging,* 675,000 USD, 2011–2016.

NASA *Hubble Space Telescope* Archival Resarch grant (AR-13250; Hogg, PI), *Probabilistic Self-Calibration of the WFC3 IR Channel*, 119,988 USD, 2013–2016.

NSF Astronomy and Astrophysics Research grant (AST-0908357; Hogg, PI), *Dynamical models from kinematic data: The Milky Way Disk and Halo,* 147,000 USD, 2009–2011.

Alexander von Humboldt Foundation Research Fellowship grant (Hogg, PI), *Cosmology with the proper motions of stars,* 32,000 EUR (approx), 2008–2011.

NASA Astrophysics Data Analysis grant (NNX08AJ48G; Hogg, PI), *Multi-wavelength astrometric catalog built from NASA data,* 277,415 USD, 2008–2011.

Amazon Web Services Research Grant (Koposov, PI) *Searching for tidal streams in the Milky Way Halo,* 40,000 CPU-hours (approx), 2009–2010.

NASA *Spitzer Space Telescope* General Observer Grant (*Spitzer* programs 50568 and 50569; Schiminovich, PI), *S5: Spitzer–SDSS Statistical Spectroscopic Survey,* 350,000 USD (approx), 2008–2010.

Google Research Grant (Blanton, PI), *Beautiful and correct SDSS images for Google Sky,* 86,000 USD, 2008–2009.

NASA Long-Term Space Astrophysics Grant (NAG5-11669; Hogg, PI), *Tools for Galaxy Astrophysics in the Era of the Space Infrared Telescope Facility,* 498,770 USD, 2002–2007.

NSF Information Technology Research Grant (AST-0428465; Hogg, PI), *Automated Astrometry for Time-Domain and Distributed Astrophysics,* 504,140 USD, 2004–2007.

NASA *Spitzer Space Telescope* General Observer Grant (*Spitzer* program 20120; Hogg, PI), *A search for PAH emission in extremely low luminosity galaxies,* 59,243 USD, 2005–2007.

NASA *Galaxy Evolution Explorer* Archival Research Grant (Blanton, PI), *K-corrections for GALEX,* 42,500 USD, 2004–2005.

NASA *Hubble Space Telescope* Archival Research Grant (Blanton, PI), *Comparing the ACS Ultra Deep Field to Low Redshift Galaxy Observations,* 70,000 USD, 2003–2004.

NSF Group Grant (PHY-0101738; Farrar, PI), *Theoretical Particle Physics, Astrophysics and Cosmology,* 686,000 USD (+16,000 USD in REU supplement), 2001–2004.

NASA Hubble Postdoctoral Fellowship, 1997–2000.

NSF Graduate Fellowship, 1992–1995.

PhDs supervised

Morad Masjedi, 2007, *Massive galaxy merging and cosmogony,* PhD thesis, New York University.

Dustin Lang, 2009, *Astrometry.net: Automatic recognition and calibration of astronomical images*, PhD thesis, University of Toronto (co-supervised by Sam Roweis at Toronto).

Ronin Wu, 2010, *Tracing star formation in the mid-infrared*, PhD thesis, New York University.

Jo Bovy, 2011, *Dynamical inference in the Milky Way*, PhD thesis, New York University.

Adi Zolotov, 2011, *The dual origin of stellar halos*, PhD thesis, New York University (co-supervised by Beth Willman at Haverford).

Tao Jiang, 2012, *Galaxy mergers and galaxy evolution*, PhD thesis, New York University.

Fengji Hou, 2014, *Bayesian inference on stellar radial velocity data*, PhD thesis, New York University (co-supervised by Jonathan Goodman at NYU).

Daniel Foreman-Mackey, 2015, *Methods for the detection and characterization of exoplanets and their population*, PhD thesis, New York University.

Mohammadjavad Vakili, 2017, *Methods in computational cosmology*, PhD thesis, New York University.

Dun Wang, 2018, *Methods for the calibration of astronomical imaging data*, PhD thesis, New York University.

Alex Malz, 2019, *Probabilistic analysis methods for cosmology using uncertainty-dominated photometric data*, PhD thesis, New York University.

Kate Storey-Fisher, 2023, *Tools for large-scale structure and galaxy formation*, PhD thesis, New York University.

Matthew Daunt, current PhD student.

Valentina Tardugno Poleo, current PhD student.

Postdoctoral scholars supervised and co-supervised

Lauren Anderson, Angus Beane, Megan Bedell, Andreas A. Berlind, Michael R. Blanton, Katie Breivik, Lisa Bugnet, Gabriella Contardo, Trevor David, Thavisha Dharmawardena, Nora Eisner, Ross Fadely, Dax Feliz, Richard Galvez, Danny Horta, Daniela Huppenkothen, Boris Leistedt, Rodrigo Luger, Melissa K. Ness, Sarah Pearson, Gabe Perez-Giz, Benjamin J. S. Pope, Adrian M. Price-Whelan, Erin Sheldon, Beth Willman, Lily L. Zhao.

292 Refereed publications

1. Hogg, D. W., Quinlan, G. D., & Tremaine, S., 1991, Dynamical limits on dark matter in the Solar System, *Astron. J.* **101** 2274–2286.
2. Hogg, D. W., Jackson, C., Żytkow, A. N., Irwin, M., Webster, R., & Tremaine, S., 1994, A photographic search for satellites of Neptune, *Icarus* **107** 304–310.
3. Hogg, D. W. & Blandford, R. D., 1994, The gravitational lens system B1422+231: Dark matter, superluminal expansion and the Hubble Constant, *Mon. Not. R. Astr. Soc.* **268** 889–893.
4. Djorgovski, S. *et al.*, 1995, Deep galaxy counts in the K band with the Keck Telescope, *Astrophys. J. Lett.* **438** L13–L16.
5. Smail, I., Hogg, D. W., Yan, L., & Cohen, J. G., 1995, Deep optical galaxy counts with the Keck Telescope, *Astrophys. J. Lett.* **449** L105–L108.
6. Smail, I., Hogg, D. W., Blandford, R., Cohen, J. G., Edge, A. C., & Djorgovski, S. G., 1995, Discovery of two giant arcs in the rich cluster A2219 with the Keck Telescope, *Mon. Not. R. Astr. Soc.* **277** 1–10.
7. Eisenhardt, P. R., Armus, L., Hogg, D. W., Soifer, B. T., Neugebauer, G., & Werner, M. W., 1996, *Hubble Space Telescope* observations of the luminous IRAS source FSC10214+4724: A gravitationally lensed infrared quasar, *Astrophys. J.* **461** 72–83.
8. Cohen, J. G., Hogg, D. W., Pahre, M. A., & Blandford, R., 1996, Strong redshift clustering of distant galaxies, *Astrophys. J. Lett.* **462** L9–L12.
9. Hogg, D. W., Blandford, R., Kundić, T., Fassnacht, C. D., & Malhotra, S., 1996, A candidate gravitational lens in the Hubble Deep Field, *Astrophys. J. Lett.* **467** L73–L75.
10. Cohen, J. G., Cowie, L. L., Hogg, D. W., Songaila, A., Blandford, R., Hu, E. M., & Shopbell, P., 1996, Redshift clustering in the Hubble Deep Field, *Astrophys. J. Lett.* **471** L5–L9.
11. Hogg, D. W., Neugebauer, G., Armus, L., Matthews, K., Pahre, M. A., Soifer, B. T., & Weinberger, A. J., 1997, Near infrared imaging of the Hubble Deep Field with the Keck Telescope, *Astron. J.* **113** 474–482. Associated erratum: *Astron. J.* **113** 2338.
12. Reid, I. N., Gizis, J. E., Cohen, J., Pahre, M. A., Hogg, D. W., Cowie, L., Hu, E., & Songaila, A., 1997, Faint M dwarfs and the structure of the Galactic disk, *Pubs. Astr. Soc. Pac.* **109** 559–565.
13. Hogg, D. W., Pahre, M. A., McCarthy, J. K., Cohen, J. G., Blandford, R., Smail, I., & Soifer, B. T., 1997, Counts and colours of faint galaxies in the U and R bands, *Mon. Not. R. Astr. Soc.* **288** 404–410.
14. Hogg, D. W. & Phinney, E. S., 1997, The fading of young stellar populations and the luminosity functions of dwarf, irregular and starburst galaxies, *Astrophys. J. Lett.* **488** L95–L99.
15. Kundić, T., Hogg, D. W., Blandford, R. D., Cohen, J. G., Lubin, L. M., & Larkin, J. E., 1997, The external shear acting on gravitational lens B1422+231, *Astron. J.* **114** 2276–2283.
16. Hogg, D. W., 1998, *On the evolution of field galaxies*, PhD thesis, California Institute of Technology.

17. Hogg, D. W. *et al.*, 1998, A blind test of photometric redshift prediction, *Astron. J.* **115** 1418–1422.
18. Hogg, D. W., & Turner, E. L., 1998, A maximum likelihood method for improving faint source flux and color estimates, *Pubs. Astr. Soc. Pac.* **110** 727–731.
19. Hogg, D. W., Cohen, J. G., Blandford, R., & Pahre, M. A., 1998, The O II luminosity density of the Universe, *Astrophys. J.* **504** 622–628.
20. Sykes, C. M. *et al.*, 1998, The complex gravitational lens system B1933+503, *Mon. Not. R. Astr. Soc.* **301** 310–314.
21. Nguyen, H. T., Eisenhardt, P. R., Werner, M. W., Goodrich, R., Hogg, D. W., Armus, L., Soifer, B. T., & Neugebauer, G., 1998, Imaging polarimetry of the gravitational lens FSC10214+4724, *Astron. J.* **117** 671–676.
22. Cohen, J. G., Blandford, R., Hogg, D. W., Pahre, M. A., & Shopbell, P. L., 1999, Caltech Faint Field Galaxy Redshift Survey. VIII. Analysis of the field J0053+1234, *Astrophys. J.* **512** 30–47.
23. Cohen, J. G., Hogg, D. W., Pahre, M. A., Blandford, R., Shopbell, P., & Richberg, K., 1999, Caltech Faint Field Galaxy Redshift Survey. VII. Data analysis techniques and redshifts in the field J0053+1234, *Astrophys. J. Suppl. Ser.* **120** 171–178.
24. Barkana, R., Blandford, R., & Hogg, D. W., 1999, A possible gravitational lens in the Hubble Deep Field South, *Astrophys. J. Lett.* **513** L91–L94.
25. Fruchter, A. S. *et al.*, 1999, *Hubble Space Telescope* and Palomar imaging of GRB 990123: Implications for the nature of gamma-ray bursts and their hosts, *Astrophys. J. Lett.* **519** L13–L16.
26. Hogg, D. W. & Fruchter, A. S., 1999, The faint-galaxy hosts of gamma-ray bursts, *Astrophys. J.* **520** 54–58.
27. Carlberg, R. G. *et al.*, 2000, Caltech Faint Galaxy Redshift Survey. XI. The merger rate to redshift 1 from kinematic pairs, *Astrophys. J. Lett.* **532** L1–L4.
28. Hogg, D. W., Pahre, M. A., Adelberger, K. L., Blandford, R., Cohen, J. G., Gautier, T. N., Jarrett, T., Neugebauer, G., & Steidel, C. C., 2000, Caltech Faint Field Galaxy Redshift Survey. IX. Source detection and photometry in the Hubble Deep Field region, *Astrophys. J. Suppl. Ser.* **127** 1–9.
29. Hogg, D. W., Neugebauer, G., Cohen, J. G., Dickinson, M. E., Djorgovski, S. G., Matthews, K., & Soifer, B. T., 2000, Three-micron imaging of the Hubble Deep Field, *Astron. J.* **119** 1519–1525.
30. Cohen, J. G., Hogg, D. W., Blandford, R., Cowie, L. L., Hu, E., Songaila, A., Shopbell, P., & Richberg, K., 2000, Caltech Faint Galaxy Redshift Survey. X. A redshift survey in the region of the Hubble Deep Field North, *Astrophys. J.* **538** 29–52.
31. van den Bergh, S., Cohen, J. G., Hogg, D. W., & Blandford, R., 2000, Caltech Faint Galaxy Redshift Survey. XIV. Galaxy morphology in the HDF (North) and its flanking fields to $z = 1.2$, *Astron. J.* **120** 2190–2205.
32. Hogg, D. W., Cohen, J. G., & Blandford, R., 2000, The Caltech Faint Galaxy Redshift Survey. XII. Clustering of galaxies, *Astrophys. J.* **545** 32–42.
33. Hogg, D. W., 2001, Confusion errors in astrometry and counterpart association, *Astron. J.* **121** 1207–1213.
34. Blanton, M. R. *et al.*, 2001, The luminosity function of galaxies in *SDSS* commissioning data, *Astron. J.* **121** 2358–2380.
35. Smette, A. *et al.*, 2001, *Hubble Space Telescope/STIS* observations of GRB 000301C: CCD imaging and *NUV MAMA* spectroscopy, *Astrophys. J.* **556** 70–76.
36. Yasuda, N. *et al.*, 2001, Galaxy number counts from the *Sloan Digital Sky Survey* commissioning data, *Astron. J.* **122** 1104–1124.
37. Hogg, D. W., Finkbeiner, D. P., Schlegel, D. J., & Gunn, J. E., 2001, A photometricity and extinction monitor at the Apache Point Observatory, *Astron. J.* **122** 2129–2138.
38. Eisenstein, D. J. *et al.*, 2001, Spectroscopic target selection for the *Sloan Digital Sky Survey*: The Luminous Red Galaxy Sample, *Astron. J.* **122** 2267–2280.
39. Stoughton, C. *et al.*, 2002, *Sloan Digital Sky Survey*: Early Data Release, *Astron. J.* **123** 485–548.
40. Schneider, D. P. *et al.*, 2002, The *Sloan Digital Sky Survey* Quasar Catalog. I. Early Data Release, *Astron. J.* **123** 567–577.

41. Hogg, D. W. *et al.*, 2002, The luminosity density of red galaxies, *Astron. J.* **124** 646–651.
42. Eisenstein, D. J., Hogg, D. W., *et al.*, 2003, Average spectra of massive galaxies in the *SDSS*, *Astrophys. J.* **585** 694–713.
43. Hogg, D. W. *et al.*, 2003, The overdensities of galaxy environments as a function of luminosity and color, *Astrophys. J. Lett.* **585** L5–L9.
44. Bernardi, M. *et al.*, 2003, Early-type galaxies in the *SDSS*. I. The sample, *Astron. J.* **125** 1817–1848.
45. Bernardi, M. *et al.*, 2003, Early-type galaxies in the *SDSS*. II. Correlations between observables, *Astron. J.* **125** 1849–1865.
46. Bernardi, M. *et al.*, 2003, Early-type galaxies in the *SDSS*. III. The fundamental plane, *Astron. J.* **125** 1866–1881.
47. Blanton, M. R., Brinkmann, J., Csabai, I., Doi, M., Eisenstein, D., Fukugita, M., Gunn, J. E., Hogg, D. W., & Schlegel, D. J., 2003, Estimating fixed-frame galaxy magnitudes in the *SDSS*, *Astron. J.* **125** 2348–2360.
48. Blanton, M. R. *et al.*, 2003, The galaxy luminosity function and luminosity density at redshift $z = 0.1$, *Astrophys. J.* **592** 819–838.
49. Blanton, M. R., Hogg, D. W., *et al.*, 2003, The broadband optical properties of galaxies with redshifts $0.02 < z < 0.2$, *Astrophys. J.* **594** 186–207.
50. Abazajian, K. *et al.*, 2003, The First Data Release of the *Sloan Digital Sky Survey*, *Astron. J.* **126** 2081–2086.
51. Hogg, D. W. *et al.*, 2004, The dependence on environment of the color–magnitude relation of galaxies, *Astrophys. J. Lett.* **601** L29–L32.
52. Quintero, A. D., Hogg, D. W., *et al.*, 2004, Selection and photometric properties of K+A galaxies, *Astrophys. J.* **602** 190–199.
53. Lupton, R., Blanton, M. R., Fekete, G., Hogg, D. W., O'Mullane, W., Szalay, A., & Wherry, N., 2004, Preparing red-green-blue images from CCD data, *Pubs. Astr. Soc. Pac.* **116** 133–137.
54. Tegmark, M. *et al.*, 2004, The three-dimensional power spectrum of galaxies from the *Sloan Digital Sky Survey*, *Astrophys. J.* **606** 702–740.
55. Tegmark, M. *et al.*, 2004, Cosmological parameters from *SDSS* and *WMAP*, *Phys. Rev. D* **69** 103501.
56. Abazajian, K. *et al.*, 2004, The Second Data Release of the *Sloan Digital Sky Survey*, *Astron. J.* **128** 502–512.
57. Finkbeiner, D. P. *et al.*, 2004, *Sloan Digital Sky Survey* imaging of low Galactic latitude fields: Technical summary and data release, *Astron. J.* **128** 2577–2592.
58. Abazajian, K. *et al.*, 2005, The Third Data Release of the *Sloan Digital Sky Survey*, *Astron. J.* **129** 1755–1759.
59. Zehavi, I., *et al.*, 2005, The intermediate-scale clustering of luminous red galaxies, *Astrophys. J.* **621** 22–31.
60. Hogg, D. W., Eisenstein, D. J., Blanton, M. R., Bahcall, N. A., Brinkmann, J., Gunn, J. E., & Schneider, D. P., 2005, Cosmic homogeneity demonstrated with luminous red galaxies, *Astrophys. J.* **624** 54–58.
61. Hogg, D. W., Tremonti, C. A., Blanton, M. R., Finkbeiner, D. P., Padmanabhan, N., Quintero, A. D., Schlegel, D. J., & Wherry, N., 2005, Mid-infrared and visible photometry of galaxies: Anomalously low polycyclic aromatic hydrocarbon emission from low-luminosity galaxies, *Astrophys. J.* **624** 162–167.
62. Blanton, M. R. *et al.*, 2005, New York University Value-Added Galaxy Catalog: A galaxy catalog based on new public surveys, *Astron. J.* **129** 2562–2578.
63. Willman, B., Blanton, M. R., West, A. A., Dalcanton, J. J, Hogg, D. W., Schneider, D. P., Wherry, N., Yanny, B., & Brinkmann, J., 2005, A new Milky Way companion: Unusual globular cluster or extreme dwarf satellite?, *Astron. J.* **129** 2692–2700.
64. Willman, B. *et al.*, 2005, A new Milky Way dwarf galaxy in Ursa Major, *Astrophys. J. Lett.* **626** L85–L88.
65. Blanton, M. R., Eisenstein, D. J., Hogg, D. W., Schlegel, D. J., & Brinkmann, J., 2005, The relationship between environment and the broad-band optical properties of galaxies in the *Sloan Digital Sky Survey*, *Astrophys. J.* **629** 143–157.
66. Hogg, D. W., Blanton, M. R., Roweis, S. T., & Johnston, K. V., 2005, Modeling complete

distributions with incomplete observations: The velocity ellipsoid from *Hipparcos* data, *Astrophys. J.* **629** 268–275.

67. Berlind, A. A., Blanton, M. R., Hogg, D. W., Weinberg, D. H., Davé, R., Eisenstein, D. J., & Katz, N., 2005, Interpreting the relationship between galaxy luminosity, color and environment, *Astrophys. J.* **629** 625–632.
68. Eisenstein, D. J., Zehavi, I., Hogg, D. W., *et al.*, 2005, Detection of the baryon acoustic peak in the large-scale correlation function of *Sloan Digital Sky Survey* Luminous Red Galaxies, *Astrophys. J.* **633** 560–574.
69. Adelman-McCarthy, J. K. *et al.*, 2006, The Fourth Data Release of the *Sloan Digital Sky Survey*, *Astrophys. J. Suppl. Ser.* **162** 38–48.
70. Farrar, G. F., Berlind, A. A., & Hogg, D. W., 2006, Foreground and source of a cluster of ultra–high-energy cosmic rays, *Astrophys. J.* **642** L89–L93.
71. Cool, R. J., Eisenstein, D. J., Hogg, D. W., Blanton, M. R., Schlegel, D. J., Brinkmann, J., Schneider, D. P., & Vanden Berk, D. E., 2006, *SDSS* pre-burst observations of recent gamma-ray burst fields, *Pubs. Astr. Soc. Pac.* **118** 733–739.
72. Masjedi, M., Hogg, D. W., *et al.*, 2006, Very small-scale clustering and merger rate of luminous red galaxies, *Astrophys. J.* **644** 54–60.
73. Blanton, M. R., Eisenstein, D. J., Hogg, D. W., & Zehavi, I. I., 2006, The scale-dependence of relative galaxy bias: Encouragement for the "halo model" description, *Astrophys. J.* **645** 977–985.
74. Tucker, D. L. *et al.*, 2006, The *Sloan Digital Sky Survey* Monitor Telescope pipeline, *Astron. Nachr.* **327** 821–843.
75. Hogg, D. W., Masjedi, M., Berlind, A. A., Blanton, M. R., Quintero, A. D., & Brinkmann, J., 2006, What triggers galaxy transformations? The environments of post-starburst galaxies, *Astrophys. J.* **650** 763–769.
76. Berlind, A. A. *et al.*, 2006, Percolation galaxy groups and clusters in the *SDSS* Redshift Survey: Identification, catalogs, and the multiplicity function, *Astrophys. J. Suppl. Ser.* **167** 1–25.
77. Tegmark, M., *et al.*, 2006, Cosmological constraints from the *SDSS* Luminous Red Galaxies, *Phys. Rev. D* **74** 123507.
78. Schneider, D. P., *et al.*, 2007, The *Sloan Digital Sky Survey* Quasar Catalog IV: Fifth Data Release, *Astron. J.* **134** 102–117.
79. Padmanabhan, N., *et al.*, 2007, The clustering of luminous red galaxies in the *Sloan Digital Sky Survey* imaging data, *Mon. Not. R. Astr. Soc.* **378** 852–872.
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Hogg, D. W. & Schiminovich, D., 2019, A better consensus: Changes to the Decadal process itself (arXiv:1907.04867).

Hogg, D. W., Price-Whelan, A. M., & Leistedt, B., 2020, Data Analysis Recipes: Products of multivariate Gaussians in Bayesian inferences (arXiv:2005.14199).

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Hainje, C. & Hogg, D. W., 2025, A formula for the area of a triangle: Useless, but explicitly in Deep Sets form (arXiv:2503.22786).

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Code and other non-standard publications

Lang, D., Hogg, D. W., Mierle, K., Blanton, M., & Roweis, S., 2008, *Astrometry.net* codebase, GPLv2 License, automated recognition and calibration for astronomical images (http://astrometry.net/).

Bovy, J., Hogg, D. W., & Roweis, S., 2009, *extreme-deconvolution* codebase, GPLv2 License, for modeling distributions in the presence of noise (http://code.google.com/p/extreme-deconvolution/).

Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J., 2012, *emcee* codebase, MIT License, an adaptive ensemble sampler (http://danfm.ca/emcee/).

Foreman-Mackey, D. & Hogg, D. W., 2012, *Daft* codebase, MIT license, for drawing graphical models (http://daft-pgm.org/).

Price-Whelan, A., Hogg, D. W., & Foreman-Mackey, D., 2017, *The Joker* codebase, MIT License, a custom Monte Carlo sampler for the two-body problem (http://github.com/adrn/thejoker).

Bedell, M., Hogg, D. W., & Foreman-Mackey, D., 2019, *wobble* codebase, MIT license, a data-driven model for making precise radial-velocity measurements (https://wobble.readthedocs.io/en/latest/).

Selected invited talks from the last three years

These are just a selection, in reverse chronological order, only since 2023-06, and only including invited talks.

Invited talk at *The Legacy and Future of Large Surveys: The Milky Way in Context*, Potsdam, Germany, 2026 May 4–8.

Physics Colloqium, Massachusetts Institute of Technology, 2020 April 16.

Astrophysics Colloquium, Stanford University and the Kavli Institute for Particle Physics and Cosmology, 2026 April 2.

Colloquium, Perimeter Institute, 2026 March 18 (cancelled for illness).

Astrophysics Colloquium, University of Toronto, 2026 March 11.

Colloquium, SkAI Institute, Chicago, 2026 January 14.

CIERA Colloquium, Center for Interdisciplinary Exploration and Research in Astrophysics, Northwestern University, 2026 January 13.

CITA Seminar, Canadian Institute for Theoretical Astrophysics, University of Toronto, 2025 October 27.

Invited talk, *Flatiron Wide Autumn Meeting*, Flatiron Institute, New York, 2025 October 15.

LEAP Seminar, Learning the Earth with Artificial Intelligence and Physics Center, Columbia University, 2025 September 18.

Königstuhl Colloquium, Max-Planck-Institut für Astronomie, Heidelberg, 2025 August 1.

Lecturer, "Scientific Afternoons" series, University of Nova Gorica, 2025 May 27.

Astrophysics Colloquium, Princeton University, 2025 March 4.

Physics Colloquium, University of Wisconsin, Milwaukee, 2025 February 28.

Courant Applied Mathematics Seminar, New York University, 2025 January 24.

Physics Colloquium, Case Western Reserve University, 2024 October 31.

Statistics Colloquium, University of Michigan, 2024 November 1.

Invited talk, *DELPHI: Deep Learning & Philosophy*, University of Graz, 2024 September 16–18.

Königstuhl Colloquium, Max-Planck-Institut fr Astronomie, Heidelberg, 2024 August 9.

Colloquium, National Optical and Infrared Astronomy Research Laboratory, 2024 March 21.

Invited talk, and panel discussion, *Symposium on the Impact of Generative AI in the Physical Sciences*, Institute for Artificial Intelligence and Fundamental Interactions, Massachusetts Institute of Technology, 2024 March 14–15.

Invited talk, Meeting in honor of Hans-Walter Rix at 60, Max-Planck-Institut für Astronomie, Heidelberg, 2024 January 19.

Physics Colloqium, State University of New York at Albany, 2023 November 17.

Physics Colloquium, University of Florida, 2023 October 19.

Invited keynote, *2023 NSF HDR Ecosystem Conference*, Colorado School of Mines, 2023 October 16–18.